Exp 4: Capture and Analyze network Traffic using wireshark.

Aim:

To capture, save, and analyse network traffic including TCP, UDP, IP, HTTP, ARP, DHCP, ICMP, ICMP, DNS protocols using the wireshark network analyzer tool.

Theory:

1) Install wireshark: sudo apt install wireshark (Linux) or download from wireshark.org.

2) Launch wireshark with admin privilages

3) Start capture by clicking the blue shark fin buttons.

4) Stop capture after ~ 30 seconds.

5) Find APR: type 'DNS' and examine query / response pairs.

6) Find DNS: 'type' icmp' and view MAC-IP resolution.

```
# Install Wireshark on Linux
sudo apt-get install wireshark -y
sudo usermod -aG wireshark $USER

# Capture with tshark (CLI Wireshark)
sudo tshark -i eth0 -c 100 -w capture.pcapng

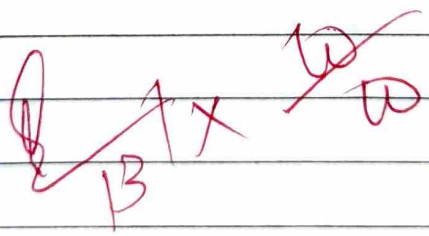
# Display filters (enter in Wireshark filter bar)
tcp                          # All TCP traffic
udp                          # All UDP traffic
http                         # HTTP only
dns                          # DNS queries/responses
arp                          # ARP traffic
icmp                         # Ping traffic
dhcp                         # DHCP DORA process
ip.addr == 8.8.8.8           # Traffic to/from specific IP
tcp.port == 443              # HTTPS traffic
http.request.method == GET   # HTTP GET requests

# tshark analysis
tshark -r capture.pcapng -Y 'http' -T fields -e ip.src -e http.host
tshark -r capture.pcapng -Y 'dns.qry.name' -T fields -e dns.qry.name
tshark -r capture.pcapng -q -z io,phs   # Protocol hierarchy stats
```

7) Filter ICMP 'type icmp' and observe ping packets

8) Export the capture file as .pcapng.

9) Use Statistics → Protocol Hierarchy to see traffic distribution.



Result:

Network traffic was successfully captured and analyzed using Wireshark.